

A Common CES 2025 Benchmark for Silicon-Sampling Models

10 items · 5 regions × 3 age groups · 150 subgroup TVs per model

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In brief. Any model, whatever it was trained on, is scored the same way: it receives the same frozen set of synthetic respondents with the same prompt, answers ten CES 2025 questions, and one script turns its answers into **150 total-variation (TV) scores** (15 subgroups × 10 items). The only differences between two runs are the model weights. This replaces the 21-subgroup protocol of September 30.

1. What is fixed and what varies

Fixed for every model	Free to vary
The 10 target items and their answer options (§2)	Base model, fine-tuning data, training recipe
The 15 subgroups and the CES 2025 human distributions (§3)	Number of training examples
The inference file: profiles and injected answers (§4)	
The prompt template (§5), temperature 1.0, one answer per call	
Parsing, scoring and bootstrap code (§6–7)	

Training rule. No CES 2025 data may be used for training or tuning, in any form. The prompt template is frozen *before* training, so a new model can be trained on exactly the format it will see at inference. Each model is delivered with a one-line manifest: base model, training sources, number of examples, and whether the anchor source survey (§4) was part of training.

2. Target items

Ten items from the CES 2025 Campaign Period Survey (prefix `cps25_`). Wording and options are taken from the codebook in the respondent’s language; the order of options is the codebook order.

#	Item	Variable	Scale
1	Spending: affordable housing	<code>spend_afford_h</code>	S
2	Spending: national childcare	<code>spend_nation_c</code>	S
3	Spending: defence	<code>spend_defence</code>	S
4	Spending: reconciliation with Indigenous Peoples	<code>spend_rec_indi</code>	S

#	Item	Variable	Scale
5	Medical assistance in dying	pos_life	A
6	Government should help build pipelines	pos_energy	A
7	Jobs before environment	pos_jobs	A
8	Number of immigrants Canada should admit	imm	I
9	Satisfaction with democracy	demsat	D
10	Personal finances over the past year	own_fin_retro	F

Answer scales:

- **S (spending):** less · about the same · more.
- **A (agreement):** strongly disagree · somewhat disagree · neither agree nor disagree · somewhat agree · strongly agree.
- **I (immigration):** more immigrants · fewer immigrants · about the same number.
- **D (democracy):** very satisfied · fairly satisfied · not very satisfied · not at all satisfied.
- **F (finances):** got better · stayed about the same · got worse.

Every scale also keeps its “don’t know / prefer not to answer” option (DK), both in the prompt and in scoring.

3. Subgroups and human distributions

Sample. All CES 2025 respondents aged 18+, living in one of the ten provinces, with a positive `cps25_weight_general_all` (w_i). No quality exclusions. Because CES 2025 is never used for training, the whole sample is evaluation data: there is no development/evaluation split.

Subgroups. 5 regions × 3 age groups = 15 cells, fixed before seeing any result and never merged.

- Region (`cps25_province`): BC · Prairies (AB, SK, MB) · ON · QC · Atlantic (NB, NL, NS, PE).
- Age (`cps25_age_in_years`): 18–34 · 35–54 · 55+.

Human distribution. For item j , cell g and answer code k , among respondents H_{jg} who were asked the item and gave any recorded answer (DK included; blank or “not asked” excluded):

$$p_{jgk} = \frac{\sum_{i \in H_{jg}} w_i \mathbf{1}(y_{ij} = k)}{\sum_{i \in H_{jg}} w_i}.$$

Cell sizes. Raw n for items asked of everyone (defence, reconciliation, immigration, democracy, finances), effective $n_{\text{eff}} = (\sum w_i)^2 / \sum w_i^2$, and the smallest item n (housing and childcare were asked of half the sample; the three agree–disagree items of a third). No cell falls below 30. The last column is the pool of Democracy Checkup 2024 respondents from which profiles are drawn (\$4).

Cell	18–34 $n / n_{\text{eff}} / \text{min}$	35–54 $n / n_{\text{eff}} / \text{min}$	55+ $n / n_{\text{eff}} / \text{min}$	DC 2024 pool (18–34 · 35–54 · 55+)
BC	600 / 493 / 193	871 / 744 / 286	1,267 / 993 / 430	316 · 353 · 490
Prairies	934 / 759 / 286	1,245 / 1,045 / 407	1,389 / 1,096 / 455	347 · 466 · 661
ON	1,918 / 1,565 / 602	2,528 / 2,163 / 799	3,191 / 2,521 / 1,035	730 · 964 · 1,300
QC	1,240 / 1,031 / 413	1,600 / 1,340 / 543	2,066 / 1,635 / 670	544 · 601 · 753
Atlantic	277 / 224 / 90	404 / 340 / 127	589 / 470 / 182	107 · 169 · 278

4. The frozen inference file

The file is built once, with a fixed seed, and handed unchanged to every model.

Profiles. For each of the 15 cells, draw **1,000 respondents** from the anchor source survey (the Democracy Checkup 2024), with replacement and with probability proportional to their survey weight. A respondent belongs to a cell through their province and their **age in 2025** (age in 2024 + 1; the survey has no one who is 18 in 2025, a negligible gap). Each profile carries, in the canonical categories used at training: age band in 2025, gender, province, education, household income and first language.

Injected answers (anchors). Each profile also carries that same real respondent’s answers to **the same seven anchor questions**, chosen from varied themes (table below). The anchors are identical for all profiles, all target items and all models. None is a target item or a close equivalent of one. Fixed anchors were preferred to random or semantically nearest questions: they give every model and every item exactly the same information, and they avoid near-duplicates that would turn the task into copying a past answer. A respondent who skipped an anchor is shown the other anchors only.

#	Theme	DC 2024 variable	Question (short)	Options
1	Ideology	dc24_lr_self_1	Left–right self-placement	0 (left) to 10 (right)
2	Partisanship	dc24_party_id	Federal party identification	Liberal, Conservative, NDP, Bloc, Green, PPC, other, none
3	Redistribution	dc24_inequality_gap	How much should be done to reduce the gap between rich and poor	Much more ... much less (5)
4	Language and identity	dc24_pos_bilingualis	We have gone too far in pushing bilingualism	Strongly agree ... strongly disagree (4)
5	Social trust	dc24_trust2	Most people can be trusted, or you need to be very careful	2 options
6	Canada–US relations	dc24_ties_us	Canada’s ties with the United States should be...	Much closer ... much more distant (5)
7	Equal rights	dc24_pos_equal_1	We have gone too far in pushing equal rights	Strongly agree ... strongly disagree (4)

All seven were asked of every DC 2024 respondent, with more than 98% substantive answers. Two further anchors (traditional family values, restrictions on religious symbols) were dropped because Azure’s fine-tuning data check flags them as hate/fairness content; so are, in training, every item scoring 4+ on Azure Content Safety’s hate scale (0–7), notably the “Canada should admit” series and several immigration and gender-role statements. Items too close to a target were ruled out: confidence in government (demsat), job creation by the private sector (jobs vs

environment), and the DC 2024 versions of immigration, democracy satisfaction and jobs vs environment. Anchors are shown with the respondent's exact option label.

Size. 15 cells × 1,000 profiles × 10 items = **150,000 calls per model**. Each profile answers all ten items, each in a separate call.

5. Prompt

The prompt is **exactly the training template** of every model we train (prompts.py in our repo), so the model sees at inference the format it learned. It exists in English and French; a profile gets the version of the language it answered the DC 2024 in, and the CES 2025 item is shown in its codebook wording in that language. The persona is a system message, the rest a user message. At inference every persona field is shown (training randomly drops some, so the model tolerates missing fields). Temperature 1.0; no retry on a completed answer (transport failures are retried). Only the chat-template wrapping (special tokens) may differ between models.

```
[system]
You are a respondent to an opinion survey conducted in 2025.
Population : Canada
Age : {age band, e.g. 25 to 34 years}
Gender : {gender}
Education : {education}
Province : {province}
Household income : {income band}
First language : {language}

[user]
Your answers to other questions in the survey:
- {anchor 1 wording} → {answer, e.g. 5 out of 10 (0 = Left, 10 = Right)}
- {anchor 2 wording} → {answer}
- ...
- {anchor 7 wording} → {answer}

Question : {CES 2025 item wording}
Options :
- {option 1}
- ...
- Don't know/ Prefer not to answer
```

Answer with the exact text of the chosen option only.

French version: *Tu es un répondant à un sondage d'opinion mené en 2025. · Tes réponses à d'autres questions du sondage : · Réponds uniquement par le texte exact de l'option choisie.*, with French field names and labels. The anchors are presented as answers to the same survey, as at training, and the survey year is 2025.

6. Scoring

Parsing. A reply is mapped to an option code by exact then normalized text match. A reply that matches no option is **invalid**. Invalid replies are removed, and the invalid rate is reported per model, item and cell. DK is a valid answer and a regular category.

Generated distribution. q_{jgk} is the share of valid replies in cell g for item j that fall on code k . All profiles have equal weight, since survey weights were applied when drawing them.

Subgroup TV (150 per model).

$$TV_{jg} = \frac{1}{2} \sum_k |p_{jgk} - q_{jgk}|.$$

TV is the share of answers that must be moved to another option to turn one distribution into the other: 0 means identical, 1 means disjoint. Zero-probability options are allowed; no smoothing.

Summaries. For each model: the 15×10 matrix; per item, the mean over the 15 cells; overall, the mean of the 150 TVs (**headline score**). As a secondary score, the same average weighted by each cell's share of CES 2025 weight.

7. Reference points and uncertainty

Every model is reported next to three references, computed once.

- **Human floor.** Each CES 2025 cell is split at random into two halves; TV between the two halves' weighted distributions, averaged over 200 random splits. It shows how much TV comes from sampling noise alone. Each half has only half the respondents, so this floor is conservative.
- **Untrained model.** The base model with the same file and prompt.
- **Modal baseline (no model).** For each item, the single most common answer in CES 2025 overall (weighted; ties go to the smallest code) is given by every profile in every cell. It ignores all subgroup differences, so a model must beat it to show it captures them. It is selected from the evaluation data itself: a diagnostic reference, not a forecast. We report how many of the 150 cells each model improves over it.

Uncertainty. 2,000 bootstrap replicates. Human respondents are resampled within cells with their weights; profiles are resampled within cells with all their answers. The same resampled profile IDs are used for every model, so differences between models are **paired**. We report basic (reverse-percentile) 95% intervals, because TV is biased upward under resampling and percentile intervals would sit above the estimate. They are reported for each model's score and for each pairwise difference. Cells with fewer than 30 respondents (raw or effective) are flagged but kept.

8. What a run delivers

The framework lives in `benchmark/` of our repo. `build` writes the frozen set once (15,000 profiles, 150,000 rendered requests, with their sha256). `run` sends it to any model through a backend (an Azure Foundry deployment or any OpenAI-compatible endpoint) and writes one row per request: `request_id`, `profile_id`, `cell`, `language`, `item`, `raw_response`, `code` (empty if invalid). `score` reads only these rows and CES 2025, and writes the 150 TVs, the summaries, the references and the paired contrasts.

9. Adding a model

Any model can join the comparison, trained by us or by anyone else, if it follows four rules.

1. **Training data.** No CES 2025 data in any form. The model comes with a short manifest: base model, training sources, number of examples, whether the DC 2024 was used.
2. **Training format.** Train on the prompt of §5 (PromptTemplate in our repo, or our training JSONL files as a model): persona as a system message; anchors, question and options as the user message; the assistant answers with the **exact text of one option**, never its number. A model trained on another format is scored on a format it has never seen.
3. **Inference settings.** The frozen messages unchanged (only the model's chat-template wrapping may differ), temperature 1.0, at most 32 output tokens, one answer per request. Reasoning modes are off (e.g. `enable_thinking=False` for Qwen3); the reply must be the option alone.
4. **Delivery.** Either (a) the model runs on its owner's machine behind an OpenAI-compatible server (Ollama, llama.cpp, vLLM, mlx_lm.server), `benchmark.run` is executed there on our frozen `requests.jsonl`, and `responses.csv` plus `run_manifest.json` are sent back — the manifest carries the sha256 of the frozen set, so we can check the requests were ours; or (b) the model is reachable at an OpenAI-compatible endpoint and we run it ourselves. We score every model with the same `benchmark.score`.